

Winding Tree

2018 HackTravel — Technical Recap

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As you can read in our [full hackathon recap](#), we had a blast during our first official Hackathon held in late October of 2018 in Prague, Czech Republic.



I was personally really happy to meet everybody, be it a crypto-enthusiast or a travel-industry veteran. And what all of this unique mixture of bright and open minds actually came up with, totally blew my mind.

Let's take a brief look at all of the 12 submissions. Bear in mind that I wasn't part of the judging process, so my evaluation is not necessarily in line with how the winners were picked.

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Air France + KLM

We are talking a lot about tackling the Airline portion of the travel industry and this team just did it in 24 hours. No more work for us then...

In all seriousness, though: They used a lot of our code and basically just replaced our off-chain storage with instances of airline APIs (be it NDC or something else) and were able to show flight search on top of Air France, KLM and Lufthansa data which was very impressive.

The resulting code is reusable to a certain extent and we took it as one of the inspiration sources for our Airline solution PoC proposal with a very interesting discussion.



Photo credits: Air France — KLM team

ATPCO

A great example of the upcoming possibilities in Swarm. With the use of both feeds and encrypted access control, this team tried to solve transfer of private information only between the owner (traveler) and a trusted party (airline) and reduce the need of data disclosure to all of the middlemen.

I was really grateful for this idea because private data encryption and transfer is one of our ultimate goals and actually one of the very few GitHub issues that are still open after a year. We will totally take some inspiration from this solution.

Photo credits: ATPCO team

Bag track

This team tried to solve the challenge proposed by Swissport — how to track bags with blockchain? Unfortunately, I am not able to find the source code, but I know I have seen it during the event itself...

The idea is to share bag events (such as *was put on a plane, landed somewhere*) with all related parties through an encrypted decentralized storage and track batched and chunked hashes of these events on the blockchain.

It's an interesting approach that might work up to a certain scale, but with the sheer load of bag events, it might be just too much data to process. Maybe Swarm feeds could be somehow used?

Air Plus

Did you know that multiple companies may actually compete together to be able to provide payment services? This project tried to solve it with a set of smart contracts that would allow you to pick the best provider possibly for

every transaction, and even without a prior written and signed agreement. Which might, unfortunately, be the problem from a legal standpoint.

But the approach seems valid to me, the smart contracts are implementing a working state machine where you can propose and accept agreements in a trustless setup which could actually be useful. The team also prepared a graphical interface in vue.js where you could interact with the smart contracts.

Securtix

Apparently, if you manage hundreds of small airlines, you have to deal with insolvency. This one-man team tried to move the settlement of insolvency insurance claims into smart contracts.

It sounds reasonable — but there are at least two caveats which were discovered during the hackathon: You would have to use trusted oracles to even know about the insolvency case And cryptocurrencies are still quite problematic for this because of high volatility.

Both of the issues can be solved though, so this might be another interesting extension to the core travel problems.

Umbral Masters

I was really hoping that somebody would pick up the challenge. The most challenging thing actually turned out to be the combination of Winding Tree's JavaScript world and NuCypher's Python ecosystem for the team.

But the direction they took is what I had in mind and I am really hopeful that we will get back to something like this in the future.

Sciant

The other team that took up the challenge with private prices, but with a totally different approach were the guys from Sciant. Interestingly, they made it as an extension of our current data model. Well done!

Managing permissions for decryption of the prices is a little clunky, but key management is always a hard problem. In this way, it potentially poses a significant amount of work for Winding Tree API operators.

The other part of their submission was taking care of booking and payment via a smart contract with all of the data (except cancellation) encrypted. That's also one of the ways we were exploring before and it's really nice to see that somebody is getting the same ideas we are.

But there are potential issues with this — namely economical. The amount of data transferred via Ethereum blockchain might become large and that means expensive. In any case, this approach (not necessarily this implementation) is one of the contenders for the Booking/Payment solution in Winding Tree.

Siesta Solution

This is big. We have a first open-source adapter for Winding Tree made by someone else than us. It's done in .NET and the guys actually successfully used it in their own real-world PMS and booking solution to create hotels and book rooms. Amazing!

It probably still needs some care, but we are very glad that the reach can be much bigger with an adapter in another technology. We would love so much to have adapters written in Python, Java and other languages. Take the challenge and contribute!

WIPU

Very neat extension of our current hotel data model was proposed by this team. You, as a hotel owner, could transparently declare partner hotels and you could pass customers to them if they have rooms available when you don't.

They worked on top of our smart contracts and actually started to wrap our stuff into a Java-based REST API. It's not functionally-equivalent to our

JavaScript APIs, but it might be useful as a starting point for somebody not really comfortable with JavaScript-based solutions.

Trabble

I'm sorry, but for some reason, I cannot even open this presentation and the linked repository is empty. But from what I remember, these guys are stitching together travel industry and chatbots. How cool would it be to book a hotel through blockchain with a chatbot?

Emergency shelter

Now that's a novel idea. It never occurred to me that we could use Winding Tree platform for such a good cause. But it is totally possible to build another platform for a very specific use case on top of it.

Technically, an entity would serve as a “shelter manager” and operate a special smart contract for that. This truly was a totally new idea to me — we can actually control the whole platform from other smart contracts. Crazy!

And we have another proof of that this idea is strong — This team extended their idea even more during the follow-up #CryptoLife hackathon by Status where they claimed one of the prizes.

Liquid Accommodation Supplements

Another possible extension of the platform might be a quick payment for “Accommodation supplements” such as laundry, bike rent etc. It's very close to the concept of amenities we use in Winding Tree and we could easily extend our data model with pricing information.

This project consisted of an Android app that would eventually use WT's amenities data and you are able to pay for them very quickly with the use of Liquidity network.

It is exactly the kind of blockchain-related services mashup that everybody should be aiming for. There are so many projects focused on particular parts

of life which you can just connect together and you can come up with something very useful.

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As you can see, there are apparently many and many use-cases in travel where blockchain or advanced cryptography have its place. And I really think we have only scratched the surface. See you next time!

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